PromoBase

Product Requirements Document

Chrome extension first release

A personal library for promo codes, with a timely reminder when a saved merchant site is visited.

Document field	Value
Version	~~0.3 - MIT license decision recorded~~ 0.4 - resulting PRD: the first release as built
Status	~~Ready for product and implementation planning~~ First release built (PromoBase 0.1.0), ready to publish
Date	8 October 2026
UPD Date	11 November 2026
Product owner	Andrii Marusiak
Primary audience	Product owner, design, engineering, quality assurance
First release	Chrome desktop extension, guest-only local storage

~~This document defines the first buildable version. It records product behavior, user-visible rules, technical choices, acceptance criteria, and future ideas so that product, design, and engineering share the same baseline.~~ This version is v0.3, the entry PRD, with the changes agreed while the first release was built (D-065). Struck-through text was dropped; text highlighted in green was added or changed, and names the decision behind it in the decisions log (docs/decisions.md).

Contents

1. Product summary
2. Problem and product opportunity
3. Goals and non-goals
4. Users and primary workflow
5. Scope and release boundaries
6. Product behavior and requirements
7. User interface and interaction rules
8. Data model and business rules
9. Technical design and stack
10. Privacy, security, and permission model
11. Market comparison and fit hypothesis
12. Success measures and validation plan
13. Risks and mitigations
14. Delivery approach
15. Future improvements
16. Decisions and assumptions
17. Glossary
18. Additional Notes
19. References

Note: Headings use Word's built-in heading styles so the document outline remains navigable in Microsoft Word and Google Docs.

How to read this version

As in v0.3, built as written	~~Dropped since v0.3~~	Added or changed (D-xxx)

1. Product summary

PromoBase is a Chrome extension that lets a person save promo codes they encounter online, organize those codes in a personal list, and receive a reminder when they visit the associated merchant website. Users enter codes manually. Creator-sponsored codes are the motivating use case, while the product remains useful for any promo code a person wants to remember.

The first release is deliberately small: no account, no server, and no sharing. Records live in the user's Chrome extension storage. ~~When the user visits a saved merchant domain, PromoBase can open its toolbar popup and show matching codes. PromoBase requests permission only for sites where the user has left Notify enabled.~~ On a site the user has saved codes for, the PromoBase toolbar icon turns yellow, with no site access at all; a click lists the codes for that site. For the sites the user chooses, PromoBase also opens its popup by itself on a visit, and asks Chrome for access to those sites only (D-061, D-064, D-100).

The product is technically feasible for a solo developer with QA experience. The key implementation risks are site-permission friction, popup interruption, and correct merchant-domain matching. The recommended stack is TypeScript with Manifest V3 for the extension, React for the interface, and Chrome extension storage. Python with FastAPI and PostgreSQL is reserved for a later account and synchronization phase.

At a glance	Initial decision
Primary job	Remember a code and surface it at the merchant site when useful.
Target platform	Google Chrome desktop, Manifest V3.
Account requirement	None for the first release.
Storage	chrome.storage.local on the current Chrome profile.
Free record limit	100 saved records.
Reminder	~~PromoBase toolbar popup, opened on a matching site visit.~~ The toolbar icon turns yellow on a site with saved codes; the PromoBase popup opens by itself on a visit to a site where the user turned reminders on (D-061, D-064).
Keeping a copy	Export to a JSON file and import it again, in the library (D-106, D-108).
First release	PromoBase 0.1.0, for Chrome 127 or later.
Market-fit hypothesis	4 out of 10 today; a focused niche with demand to validate.

2. Problem and product opportunity

2.1 User problem

People hear or see promo codes in creator videos, streams, podcasts, newsletters, and other online content. They may not need the advertised product at that moment. A note in a separate app can be forgotten, and the user has to remember which merchant the code belongs to when they eventually shop.

2.2 Product opportunity

PromoBase provides a lightweight personal memory layer. The user records a code and its merchant once. Later, the extension recognizes the merchant site and presents the saved code without requiring the user to search through notes or remember the code. The user remains in control of which merchant sites PromoBase can access.

2.3 Product positioning

PromoBase is a personal promo-code organizer with context-aware reminders. It is not a coupon discovery engine, affiliate shopping portal, price-comparison tool, or automatic checkout optimizer. Its initial value comes from remembering codes the user already found.

3. Goals and non-goals

3.1 First-release goals

- Let a guest add, view, edit, and delete their own promo-code records without registration.
- Make saving a code quick enough to use while watching a video or browsing another page.
- Publish the source code in a public GitHub repository with a clearly selected open-source license and safe contribution guidance.
- Match a saved merchant registrable domain across its root hostname and subdomains.
- Open a PromoBase toolbar popup when the user visits a matching site and has allowed the required site access. Without any site access, mark the toolbar icon on sites with saved codes (D-061).
- Keep expired codes discoverable, clearly marked, and lower in the list rather than deleting them.
- Explain local data loss plainly and keep the guest limit visible.

3.2 Non-goals for the first release

- Creating an account, cloud backup, cross-device sync, or recovery after removing the extension. A file exported from the library is the way to keep a copy (D-106).
- Sharing codes with other users or receiving shared codes.
- Finding, scraping, validating, testing, or auto-applying codes from the web.
- Reading or analyzing video content, creator descriptions, browsing history, or page text beyond the minimum URL needed for a configured reminder.
- Subscription billing, embedded advertisements, donations, affiliate tracking, or in-extension feedback submission.
- Native mobile apps, Firefox, Edge, Safari, or Chrome mobile support.

4. Users and primary workflow

4.1 Primary user

A Chrome user who shops online and wants to save discount codes found from creators or elsewhere, without opening an account. The initial product owner is also the first likely power user and quality reviewer.

4.2 Primary workflow

1. The user hears a code such as QWERTY in a video and opens PromoBase from the Chrome toolbar.
2. The user enters the code and, when available, the merchant URL or domain, any dates or note. ~~Notify is checked by default only when a usable merchant URL is present.~~ Only the code is required; the popup opens on its Add tab when the current site has no saved codes (D-081, D-092).
3. PromoBase validates the fields, derives the merchant registrable domain, saves the record locally~~, and requests site access if required for reminders.~~, and offers to remind the user on that site. Chrome asks for access to the site only if the user accepts, after PromoBase has explained why (D-061, D-064).

4. Later, the user visits the merchant website. ~~PromoBase finds eligible saved records for that merchant and opens its toolbar popup.~~ The PromoBase icon turns yellow, and a click shows the codes saved for that site (D-093, D-094). If reminders are on for the site, the popup opens by itself, once per visit (D-100, D-102).
5. The popup displays each matching code, validity context, and an action to copy the code. The user can dismiss the popup and continue browsing.

4.3 Example record

Field	Example value	Behavior
Promo Code	QWERTY	Required. Trimmed text value. Up to 100 characters (D-073).
Resource or Product URL	https://www.dropbox.com/	Optional. Used to derive the merchant domain. Up to 2,000 characters. The derived site is shown before saving and can be corrected (D-017, D-073).
Start Date	Not set	Optional. If set, reminders begin on that local calendar date.
Expiry Date	31.12.2026	Optional. Valid through the end of that date in the current timezone.
Note	2 TB plan discount	Optional. Maximum 140 characters.
Notify	~~Enabled when a usable merchant URL is supplied~~ Reminders on dropbox.com	~~Checked by default only when a usable URL is present. Otherwise off and disabled.~~ Off by default. After a save, PromoBase offers to turn reminders on for the site; the switch applies to every code for that site (D-064, D-100).

5. Scope and release boundaries

5.1 Included in the first release

- Guest onboarding with an always-visible data-retention warning. A welcome opens in a new tab when PromoBase is installed (D-113).
- A local list with a 100-record maximum, record count, search, and active/expired ordering.
- Create, edit, and delete operations for code records.
- Manual entry and validation of all record fields.
- Merchant domain normalization and reminder matching across subdomains.
- A permission-aware toolbar popup reminder with code copying and expiry warnings. A toolbar icon that turns yellow on sites with saved codes, with no site access (D-093).
- Current-timezone date display and local calendar-day date semantics.
- Accessible labels, keyboard-operable controls, and clear empty/error states.
- Public GitHub repository with a selected license, setup instructions, and no secrets or real user data.
- Export of the saved codes to a JSON file, and import of such a file (D-106, D-108).
- A privacy policy, the Chrome Web Store listing text, icon, tile and screenshots (D-114, D-116, D-117).
- Automated checks of the built extension in Chrome, run in CI, and a release package built from a tag (D-121, D-125, D-126).

5.2 Explicitly deferred

Accounts, cloud storage, sharing, monetization, and donations remain future options. They are not prerequisites for learning whether the save-and-remind behavior is useful.

6. Product behavior and requirements

The identifiers below are stable references for implementation tasks and later acceptance testing. All requirements in this section define the initial release scope unless explicitly deferred elsewhere in this document.

PB-001 Guest access

A user can start using PromoBase without creating an account.

Acceptance criteria

- First launch opens a concise welcome view with a clear Add a code action. It opens in a new tab on install only, not on an update or a reload; it says that PromoBase does not check whether a code works, and asks the user to pin PromoBase while it is not on the toolbar (D-113).
- No email address, password, or sign-in step is required to create or manage local records.
- ~~The interface identifies the current storage mode as guest and local.~~ The interface says where codes are kept ("Codes stay in this Chrome profile."). "Guest mode" is not shown while it is the only mode (D-128).

PB-002 Data-retention warning

A persistent warning explains where guest records are stored and what can remove them.

Acceptance criteria

- The warning is visible in the main extension interface and remains available in settings or an About view. The full notice is at the foot of the popup and in the library's About section; the library's heading says where codes stay.
- ~~Use this plain-language wording or an equivalent: Your promo codes are saved in this Chrome profile. Clearing browsing history will not remove them. If you remove PromoBase or reset or lose this Chrome profile, your saved codes may be lost. This version does not include cloud backup.~~ Wording: "Your codes stay in this Chrome profile when you clear browsing history, but are lost if you remove PromoBase or reset or lose the profile. Keep a copy with Export in the library." The last sentence is left out while saved codes cannot be read (D-089, D-107, D-124, D-128).
- Do not claim that clearing Chrome browsing data clears extension records. Chrome documents extension storage as persisting through browsing-data clearing and being cleared when the extension is removed [R1, R2].

PB-003 Record capacity

Guest users can store up to 100 records.

Acceptance criteria

- The list shows the number of saved records and the 100-record limit.
- The 100th record can be saved. The next save is blocked with a clear explanation and next step. At the limit, Add a code is disabled and its tip says why; an import skips the codes past the limit and names them (D-108, D-111).
- Editing an existing record remains possible at the limit.
- Deleting a record releases one slot.

PB-004 Create a record

The user can enter and save a promo-code record with the fields listed in Section 8.

Acceptance criteria

- Promo Code is required and cannot be blank after trimming leading and trailing whitespace. Up to 100 characters (D-073).
- Resource or Product URL, Start Date, Expiry Date, and Note are optional. The address is limited to 2,000 characters (D-073).
- ~~When a valid Resource or Product URL is present, Notify is checked by default for a new record.~~ A new record has reminders off. After a save with a usable site, PromoBase offers to turn reminders on for that site (D-064). A record saved for a site that has reminders on gets them too (D-100).
- When the Resource or Product URL is empty, ~~Notify is off and disabled.~~ The record can still be saved, but cannot trigger a site-visit reminder.
- If a user removes the URL while editing a record, ~~disable Notify and turn it off~~ its reminders go off; remove reminder access if no other enabled record needs it. An edit never turns reminders on by itself; a record moved to another site follows that site's switch (D-075, D-100).
- Note is limited to 140 characters and the interface communicates the remaining or used character count.
- Invalid URLs and invalid date combinations receive actionable inline messages.
- A duplicate (the same code, ignoring case, for the same site) and an already expired code are saved with a warning, never blocked (D-020, D-077).
- Unfinished input in the popup is kept when the popup closes and restored when it opens again, until it is saved or cleared (D-082).

PB-005 Merchant domain normalization

PromoBase converts an entered URL or hostname to a canonical registrable domain for matching.

Acceptance criteria

- Accept a full URL such as https://www.dropbox.com/plans, or a domain such as dropbox.com.
- Ignore scheme, path, query, fragment, port, and a leading www when deriving the matching key.
- www.dropbox.com, dropbox.com, and shop.dropbox.com resolve to the same saved merchant key dropbox.com.
- Match on DNS label boundaries. dropbox.com.example.org and notdropbox.com do not match dropbox.com.
- Use Public Suffix List-aware registrable-domain parsing so suffixes such as co.uk are handled correctly [R13]. The private section of the list is included, so shop.myshopify.com is a site of its own (D-016, D-071).
- Refused, each with its own message: text that is not a web address, schemes other than http and https, an address with a user part before @ (dropbox.com@evil.example), IP addresses, localhost and one-word names, and a shared suffix on its own such as co.uk or github.io (D-018, D-069).
- The derived site is shown before saving, and the user can correct it (D-017, D-078).
- International names are stored in their xn-- form and shown in their readable spelling with the stored one next to it, so a look-alike name cannot pass for another site (D-086).
- A saved site is checked for its spelling only when read; an update of the suffix list never changes or invalidates a saved one (D-070).

PB-006 Site permission

PromoBase requests access only for saved merchant domains where ~~Notify is enabled~~ the user turned reminders on.

Acceptance criteria

- The permission explanation states that PromoBase checks visits to the saved site only to show matching saved codes. PromoBase shows its own explanation before Chrome's dialog (D-061).
- The permission request is initiated from a clear user action, ~~such as saving a record with Notify checked~~ turning reminders on for a site, consistent with Chrome's runtime permission model [R4]. The request is the last thing that click does, after everything is saved, because Chrome closes

the popup when its dialog appears (D-057, D-058). Several sites waiting for access can be allowed in one dialog (D-109).

- If permission is declined, the record is still saved and can be opened from the extension, while the record indicates that automatic reminders need site access. Whether access is held is read from Chrome every time, never stored (D-014).
- The user can later enable access and reminders from the record or settings. Both the popup and the library offer it.
- Turning Notify off removes PromoBase's need to monitor that merchant for reminders ~~and should unregister its content script or access where Chrome permits~~. Access a site no longer needs is given back, and access in any other shape is never touched (D-103). Chrome keeps listing a site the user once allowed on its own Site access page; no extension API clears that list (D-063, D-101).

PB-007 Visit reminder

When the user reaches a saved merchant site, PromoBase opens its toolbar popup with matching codes.

Acceptance criteria

- Reminder is triggered on a top-level visit to a matching HTTP or HTTPS registrable domain for which ~~Notify~~ reminders and site access are both enabled. Visits are seen from tab updates, with no script in the page (D-049). The popup opens only when the tab is active and its window focused; otherwise the tab gets a badge and the popup opens when the user comes back to it (D-001, D-002, D-050, D-051).
- A single popup groups all matching eligible records rather than opening one popup per code.
- The popup shows the promo code, merchant, note when present, start/expiry context, and a Copy code action.
- When multiple records match, show all of them in one popup and order non-expired records before expired records.
- Show at most one automatic popup per registrable domain visit per tab. Do not reopen on reloads or same-domain route changes; allow a new reminder after the user leaves that registrable domain and later returns. A site also waits 10 minutes before reminding again in any tab, and its subdomains count as one visit (D-003, D-004, D-102).
- If a popup cannot be opened or the site permission is unavailable, do not lose the record. Keep it accessible from the extension action and use an unobtrusive badge where practical. Tabs restored when Chrome starts get a badge, not a popup (D-060).
- Chrome exposes action.openPopup from Chrome 127 onward; this is the implementation basis for a toolbar popup reminder [R5, R6].
- "Don't remind me again" in the reminder turns reminders off for the whole site (D-005, D-100).
- Codes that have not started are left out of the automatic popup, and a site with only such codes opens none (D-022, D-104).
- On every site with saved codes, whether reminders are on or not, the toolbar icon turns yellow without any site access, and a click on it lists the codes for that site [R16, R17] (D-008, D-061, D-093, D-094).

PB-008 Validity and expiry

Expiry affects display and warning text but does not silently remove records or suppress visit reminders.

Acceptance criteria

- A code with no start or expiry date is treated as valid indefinitely for list ordering and reminders.
- A future Start Date prevents reminders until the beginning of that date in the current local timezone. Before then, show the record as Upcoming in the list.
- An Expiry Date is inclusive through 23:59:59.999 on that local calendar day. Starting the next day, mark the code Expired.
- Expired records remain visible, appear gray or otherwise visually subdued, and sort after non-expired records. They are outlined instead of filled (D-083).

- Visiting the merchant for an expired code still opens the reminder and labels it clearly, for example: ~~This code may have expired. Its saved expiry date was 31.12.2026.~~ Expired 31.12.2026.
- If Start Date is later than Expiry Date, block save and explain the conflict. The same day for both is allowed (D-074).

PB-009 List and find

The user can review saved records and find them by code or merchant.

Acceptance criteria

- Show merchant, code, date status, note preview, and Notify state in each row or card. The library groups records under Active, Upcoming and Expired with counts, and tags only Upcoming and Expired rows (D-083).
- Search is case-insensitive and matches code, normalized merchant domain, and note. An international name is found by either spelling (D-086).
- Sort active and undated records first, upcoming records next, and expired records last. Within a status group, newest records appear first.
- An empty state explains how to add the first code. A no-results state offers a way to clear search.
- The popup has three tabs: This site (the codes for the current site), All codes (a compact list with search and Copy) and Add (D-092, D-094).

PB-010 Edit a record

The user can change any saved field without changing the record identity.

Acceptance criteria

- Opening edit shows the current saved values.
- The same validation and date rules as "create" apply.
- Changes are saved to local storage and reflected immediately in the list and reminder registration. An edit keeps the saved address and site exactly as stored unless the user changes one of them (D-078).

PB-011 Delete a record

The user can delete a record they no longer need.

Acceptance criteria

- Delete requires a confirmation step or a reversible undo affordance. Delete asks in place, with the focus on Keep; there is no undo (D-083).
- After deletion, the record no longer appears in list results or future reminders.
- Its site registration or permission is removed only if no remaining ~~Notify-enabled~~ reminder-enabled record needs that merchant domain.

PB-012 Copy code

The user can copy a displayed code with one action.

Acceptance criteria

- Copy uses the browser clipboard when available and displays a short success message.
- If copying fails, the code remains selectable and a useful fallback message is shown.

PB-013 Local persistence

Records are saved using Chrome extension storage and survive ordinary browser restarts and browsing-data clearing.

Acceptance criteria

- Use chrome.storage.local rather than localStorage ~~in a content script~~.

- Records remain after closing and reopening Chrome and after clearing ordinary browsing data.
- Records are expected to be removed when PromoBase is uninstalled or the Chrome profile is reset or lost [R1, R2].
- The initial release does not promise cross-device sync, backup, or recovery. Export and import (PB-016) let the user keep a copy by hand.
- One storage key holds { schemaVersion, records }. Every change is a read, change and write under one lock, through one module, so pages and the service worker cannot overwrite each other (D-041, D-053, D-076).
- Data that cannot be read is reported and never repaired or overwritten (D-042). A change to the stored layout follows a checklist with a version bump, a migration and frozen test files (D-119).

PB-014 Save and error recovery

PromoBase explains failures clearly and protects the user's unsaved work when an operation cannot complete.

Acceptance criteria

- Do not show a success state unless the storage operation has completed successfully.
- If a create or edit save fails, keep the entered values on screen and show a plain-language message with a retry action where retry is safe.
- If Chrome storage is unavailable or rejects a write, do not clear the form or imply that the record was saved. Keep existing saved records visible when possible.
- If validation fails, identify the field and explain what the user can correct. Do not treat a missing optional URL as a save error.
- If site permission is denied or cannot be requested, save the record if local storage succeeds, preserve the user's ~~Notify selection~~ reminder choice, and explain that reminders are unavailable until access is enabled.
- If a matching-site popup cannot open, preserve the record and provide access from the toolbar action or library; do not repeatedly retry during the same visit.
- Do not transmit error details, promo codes, notes, or visited URLs to a server in the first release.
- Saved codes this version cannot read are not a dead end: the library offers a copy of the stored data exactly as stored, and a way to report the problem (D-124).
- A failure message stays until the next change or until dismissed; a confirmation goes away after 10 seconds (D-090, D-112).

PB-015 Public source repository

PromoBase source code is maintained in a public GitHub repository under an explicit open-source license.

Acceptance criteria

- Before public release, add an MIT license file and identify MIT in the README.
- README explains the project purpose, supported Chrome version, local data behavior, installation for development, and how to report issues.
- Do not commit credentials, signing keys, private test data, or real user promo-code records. Use synthetic examples in documentation and tests.
- Review third-party dependency licenses before distribution and keep the Chrome Web Store privacy policy and data-use disclosures accurate [R3, R14~~, R15~~]. The extension ships THIRD_PARTY_NOTICES.txt, made at build time from the packages it bundles (D-122). The privacy policy is PRIVACY.md and the listing text store/listing.md (D-116).
- The repository holds CONTRIBUTING.md, RELEASING.md and docs/ with this PRD and the decisions log (D-119, D-120, D-121).

PB-016 Export and import

The user can keep a copy of the saved codes in a file and bring it back, in this profile or another (D-027, D-106, D-108).

Acceptance criteria

- Export downloads promobase-codes-YYYY-MM-DD-HH-MM.json with the saved records, without any extra permission (D-106).
- Import takes a PromoBase export. The whole file is checked before anything is saved and refused whole, with the position of the code that fails; a file over 2 MB is refused before it is read (D-108).
- An import only adds: a code saved already is skipped, codes past the 100 limit are skipped, and a duplicate is flagged. Every skipped or flagged code is named in a report that stays until dismissed (D-108).
- A site already in the library keeps its reminder switch; a new site takes the file's. Sites that then need access can be allowed in one dialog (D-108, D-109).

7. User interface and interaction rules

7.1 Primary surfaces

Surface	Purpose	Key content
Toolbar popup	Quick add and entry point to the library.	~~Add code, search/list summary, record limit, local-data warning.~~ Three tabs, This site, All codes and Add; it opens on This site when the current site has saved codes, otherwise on Add. Record count and limit, local-data notice, link to the library (D-007, D-081, D-092, D-094).
Full library view	Review and maintain saved records. A page in its own tab (D-024).	Search, status grouping, edit, delete, and reminder permissions. Export, Import, the welcome when empty, About your saved codes (D-106, D-108, D-113).
Automatic reminder popup	Surface codes relevant to the current merchant.	Matching codes, copy actions, date status, explicit expired warning. "Don't remind me again" (D-104).
Permission explanation	Explain site access at the point ~~Notify is enabled~~ the user turns reminders on for a site.	Merchant domain, why access is needed, and how to proceed if declined.
Toolbar icon	Show that the current site has saved codes, with no site access (D-061).	The ticket turns yellow on such a site and returns to its usual look elsewhere (D-093, D-114).

The popup stays under Chrome's 600px height limit in its normal states, with room for wider system fonts; when a panel has more than fits, only that panel scrolls and the tabs, the count and the notice stay in view (D-084, D-095, D-127, D-128).

7.2 Interaction principles

- Keep the "save" flow compact. Required fields should fit in the initial view, with optional fields visible without making the task feel heavy.
- Do not interrupt a site visit with a generic permission prompt. Explain the matching behavior before the user grants access.
- Show expired status in both the library and the reminder popup. Never present an expired code as confirmed valid.

- Use clear labels such as Upcoming, Active, and Expired rather than relying on color alone.
- Support keyboard focus, visible focus indicators, readable contrast, and labels that screen readers can announce. An accessibility pass found no axe-core violations in 16 states of the popup and the library, light and dark, and field outlines reach at least 3:1 contrast (D-115).
- Dates are typed as DD.MM.YYYY in a masked field, with no calendar; a two-digit year is completed to 20YY when the field is left (D-033, D-085).
- Import, Export and Add a code have a one-sentence tip, shown after a short rest of the pointer or at once on keyboard focus (D-111).

7.3 Repeat reminder policy

The initial behavior is one automatic reminder per registrable-domain visit per tab. Refreshes and same-domain navigation should not reopen it. Leaving the merchant domain and returning starts a new visit. This balances visibility with avoiding repeated interruptions. A site also waits 10 minutes before it reminds again in any tab; the wait resets when its reminders are switched on or off. Tabs restored when Chrome starts get a badge instead of a popup (D-003, D-004, D-060).

8. Data model and business rules

8.1 Record fields

Field	Type	Required	Rule
id	UUID string	Yes	Stable record identifier generated on creation.
promoCode	String	Yes	Trimmed, non-empty text. Preserve case for display and copy. Up to 100 characters (D-073).
resourceUrl	String	No	Optional source URL entered by the user. Up to 2,000 characters (D-073).
merchantDomain	String	Derived	Canonical registrable domain, for example dropbox.com. Or the user's correction of it. Read back checking its spelling only (D-017, D-070).
startDate	Date string (ISO)	No	Stored YYYY-MM-DD local calendar date; shown as DD.MM.YYYY.
expiryDate	Date string (ISO)	No	Stored YYYY-MM-DD; shown as DD.MM.YYYY; inclusive through end of local day.
note	String	No	Maximum 140 characters.
notify	Boolean	Yes	~~Defaults to true only when a usable merchant domain is available; otherwise false.~~ Whether reminders are on for the record's site: false for a new record, set for every record of a site by its switch (D-064, D-100). Must be false when no usable merchant domain exists.
createdAt	UTC timestamp	Yes	Creation time for deterministic ordering and troubleshooting.
updatedAt	UTC timestamp	Yes	Updated on every successful edit.
schemaVersion	Number	Yes	Stored once, next to the records ({ schemaVersion, records }); 1 in the first release (D-023, D-041).

8.2 Date representation

Start Date and Expiry Date are calendar dates, not clock instants. In user-facing forms and lists, display and accept dates as DD.MM.YYYY, for example 31.12.2026. Store them internally as ISO date strings such as 2026-12-31 so comparisons and sorting are reliable. Compare them with the user's current local calendar date. This preserves the agreed rule that an expiry is valid through the end of that day in the current timezone, without daylight-saving changes moving the cutoff. Use UTC timestamps only for events such as createdAt and updatedAt. Entry accepts real calendar days from 2000 to 2100 (D-021, D-074).

8.3 Domain normalization examples

User input or visited host	Canonical key	Match
https://www.dropbox.com/plans	dropbox.com	Yes
dropbox.com	dropbox.com	Yes
shop.dropbox.com	dropbox.com	Yes
dropbox.co.uk	dropbox.co.uk	Only with same registrable domain
notdropbox.com	notdropbox.com	No
dropbox.com.example.org	example.org	No
shop.myshopify.com	shop.myshopify.com	Only that shop (D-016)
münchen.de	xn--mnchen-3ya.de	Yes; shown as münchen.de (D-086)
192.168.0.1, localhost, co.uk	None	Refused when saving (D-069)

8.4 Example TypeScript shape

```
type PromoCodeRecord = {
  id: string;
  promoCode: string;
  resourceUrl?: string;
  merchantDomain?: string;
  startDate?: string;   // YYYY-MM-DD, local calendar date
  expiryDate?: string;  // YYYY-MM-DD, inclusive
  note?: string;        // 140 characters maximum
  notify: boolean;
  createdAt: string;    // ISO UTC timestamp
  updatedAt: string;    // ISO UTC timestamp
};
```

The merchantDomain field is derived when a valid resource URL or bare domain is provided. A record without a usable merchant domain can still be stored and managed, but it cannot generate a site-visit reminder until the user adds one. The shape above is the one built (src/domain/record.ts). Absent fields are left out rather than stored empty.

9. Technical design and stack

9.1 Recommended first-release stack

Layer	Technology	Why it fits
Browser extension	Chrome Manifest V3	Current Chrome extension model, event-driven service worker, toolbar action, optional host permissions.
Language	TypeScript	Strong types for extension APIs, record validation, and maintainable UI state.
UI	React with Vite (React 19, Vite 8, no extension framework, D-038)	Familiar component model and fast local development. Keep the UI small and avoid heavy dependencies.
Local records	chrome.storage.local	Persistent extension-specific storage with ample quota for 100 short records [R1].
Domain parsing	tldts, which bundles the Public Suffix List (D-071)	Loaded only by the pages that save codes, never by the service worker (D-096).
Site detection	~~Optional host access plus dynamically registered content script~~ Optional host access per site; tab updates; declarativeContent for the icon	~~Inject only on merchant domains explicitly enabled for reminders [R4, R7].~~ No script runs in any page. With access, the tab's address shows a visit; without it, Chrome itself changes the icon on saved sites [R4, R16] (D-049, D-061).
Reminder UI	chrome.action.openPopup	Opens the PromoBase toolbar popup on a match in Chrome 127 and later [R5, R6].
Account backend	None in first release	Avoids authentication, hosting, and data-protection overhead until local utility is validated.
Later account backend	Python FastAPI and PostgreSQL	Uses Andrii's Python strength; relational records and ownership fit sync and sharing.
Checks	Vitest, ESLint, Prettier; browser checks in Chrome for Testing	Unit tests for every logic module, and the built extension driven in a real Chrome in CI (D-030, D-125, D-126).

9.2 Runtime flow

1. The user saves a record ~~with Notify enabled~~, then turns reminders on for its site.
2. The extension explains the requested merchant access and asks Chrome for the optional host permission during the user's ~~save action~~ click, as the last thing the click does (D-057, D-058).
3. ~~If granted, the extension registers a minimal content script for that registrable domain. The script reads the page URL only and sends the domain and tab context to the service worker.~~ If granted, Chrome tells the service worker about tab updates on that site, with the tab's address and nothing from the page (D-049).
4. The service worker loads matching records from chrome.storage.local, applies start-date and repeat-reminder rules, then opens the action popup. The decision is a pure function of the visit state and what Chrome reports, checked by unit tests (D-102).
5. The popup presents the matching record data and copy controls. It does not upload the record or browsing URL to a server.

Separately, the service worker keeps one declarativeContent rule listing every saved site, so Chrome changes the toolbar icon on those sites without telling PromoBase which page is open (D-093).

9.3 Permission baseline

Expected baseline permissions are storage, ~~scripting,~~ activeTab, declarativeContent and optional host access for supported HTTP and HTTPS merchant sites, granted one site at a time as *://*.<site>/* (D-008, D-043, D-061). ~~The exact manifest must be confirmed during implementation.~~ The manifest is pinned by a test [R11]. Avoid broad access to all websites, browsing history, cookies, or tabs unless a demonstrated feature requires it. Chrome recommends the narrowest permissions needed and supports optional permissions requested at runtime [R4].

9.4 Later account phase

When account-backed persistence is justified, add an email-based identity flow, FastAPI service, and PostgreSQL database. The extension remains the client. The server owns users and records, enforces record ownership, and later supports sync and sharing. Use HTTPS for all requests, short-lived access tokens, secure refresh handling, and account deletion/export flows. Do not store user passwords in extension storage. MongoDB is not recommended for this data model unless requirements materially change.

9.5 Open-source project

PromoBase is intended to be developed in a public GitHub repository. This does not change the local-first data model: a public source repository does not publish users' locally stored promo codes. Before making the repository public, remove credentials, private test data, and personal user records; use synthetic examples only. MIT is the selected open-source license. Add the MIT license file to the repository and identify it in the README before public release. GitHub notes that a public repository without a license remains subject to default copyright restrictions, even though other GitHub users may view and fork it [R14]. Keep dependency licenses and Chrome Web Store privacy disclosures accurate. If the shipped extension handles user data, publish an up-to-date privacy policy describing its collection, use, and sharing [R3~~, R15~~]. Built: the repository is public under MIT at github.com/amarusyak/PromoBase, with a privacy policy, licence notices, contribution and release guides, and dependency updates by Dependabot (D-037, D-116, D-121, D-122, D-129).

10. Privacy, security, and permission model

- Local-first: do not send promo codes, notes, domains, or browsing activity to a PromoBase server in the first release.
- Least privilege: ask for site access only for domains with ~~Notify enabled~~ reminders on; explain that access is used to show saved codes on visits.
- Minimal page access: ~~the content script needs the current hostname and~~ PromoBase runs no script in any page; with access it reads only the tab's address, and it must not inspect page text, forms, cart contents, passwords, or checkout information.
- No analytics by default: avoid collection of browser history and user behavior in the initial version. Product learning can use voluntary interviews and manual feedback.
- Data retention: local records persist through ordinary browsing-data clearing, but extension removal or Chrome profile loss may delete them [R1, R2].
- Store listing: before Chrome Web Store publication, provide a privacy policy and accurate data-use and permission disclosures if any data handling changes or the store requires disclosures for the shipped feature set [R3]. PRIVACY.md and store/listing.md, kept in step with the manifest by tests; data handled only in the browser is still disclosed (D-116).
- Open or restricted pages: Chrome internal pages and some protected pages cannot host extension scripts. The UI should explain that reminders only work on supported web pages. The popup says when the current tab is not a web page (D-094).
- Incognito: off until the user allows it in Chrome; then the same codes serve normal and Incognito windows, and what PromoBase notes about an Incognito tab goes when the tab closes (D-123).
- The extension makes no network requests and loads no remote fonts or scripts.

11. Market comparison and fit hypothesis

PromoBase participates in the broad coupon-extension market but solves a narrower memory problem. The products below are compared by their publicly described core behavior, not by a full feature audit.

Product	Core behavior	Overlap with PromoBase	PromoBase distinction
PayPal Honey	Searches for and tests available coupon codes at checkout; may apply a working code [R8].	Displays code opportunities while the user shops.	PromoBase stores codes the user already encountered, including creator-specific codes, and remembers them across time.
Capital One Shopping	Tests community-sourced codes, compares prices, and offers shopping rewards [R9].	Shows relevant promo opportunities on merchant sites.	PromoBase is a personal saved-code library, not an automatic discovery, price, or rewards service.
Rakuten	Alerts users to cashback, coupons, and deals and supports coupon application at supported stores [R10].	Uses browser extension prompts during shopping.	PromoBase has no affiliate attribution or cashback activation. It surfaces the user's own saved records.
PromoBase first release	Manual save, local organization, and domain-based reminder.	Shares the general goal of helping users remember discounts.	Creator-to-code memory, user-owned records, local-first behavior, and no account requirement.

11.1 Fit score

Current market-fit hypothesis: 4 out of 10. This is a qualitative product judgment, not a measured demand statistic. Coupon use is established, but PromoBase asks users to manually save codes and competes for attention with mature extensions that automate coupon discovery. Its clearest potential segment is people who frequently hear creator-sponsored codes and later shop on the advertised sites.

The concept is worth prototyping because the first version is inexpensive to build and directly tests a specific user habit. A higher score should require evidence that users save codes repeatedly, return to the reminders, and prefer this workflow over notes, bookmarks, or existing coupon extensions.

11.2 Positioning statement

For people who encounter promo codes before they are ready to shop, PromoBase is a personal code library that remembers the merchant and brings the code back when they visit that site. Unlike automatic coupon engines, PromoBase focuses on the codes the user chose to save.

11.3 Build viability

Estimated build viability for a solo developer: 8 out of 10 for the local first release. The core involves standard extension UI, local record management, date rules, domain normalization, and a narrow permission-aware reminder flow. The main learning curve is Chrome extension lifecycle and permission behavior rather than record-management logic. The full account, sync, sharing, and payment vision is a larger product and operations commitment, estimated at 5 out of 10 for solo delivery without reducing scope or using managed services.

12. Success measures and validation plan

The first release should prioritize observed usefulness and reliability over growth or revenue. Because the MVP is local-first and has no analytics, measure through opt-in user interviews, task observations, and local development checks rather than hidden tracking.

Measure	Initial signal	How to assess
Save-flow completion	A new user can save a valid code without assistance.	Observe 5 to 10 first-time users and note errors or abandonment.
Reminder usefulness	Users recognize a matching merchant prompt and can copy the code.	Ask users to save a test code, revisit the domain, and describe whether timing felt helpful.
Permission acceptance	Users understand why one merchant site needs access.	Record consent/decline in a voluntary test session, not by uploading telemetry.
Repeat use	Users save multiple records over several weeks.	Follow up with volunteers and ask how often they used or ignored the extension.
Reliability	No record loss, false domain match, or repeated-popup defect in tested flows.	QA test matrix across supported Chrome versions and date/timezone scenarios. Unit tests and the browser checks run on every change to main, in a pinned Chrome for Testing; Chrome 127, the minimum, is not yet run automatically (D-125, D-126).

12.1 Early validation questions

- How often do people encounter codes before they intend to buy?
- Would they save a code manually, or is that extra step too much friction?
- Is a toolbar popup helpful or disruptive when arriving at a merchant site?
- Will users grant site access when the purpose is limited and clearly explained?
- Do the creator name, source video URL, or product category need to be captured in the first version? Not in the first release; the note field carries it.
- Is the yellow icon enough on its own, or do users turn reminders on? The answer decides how much the automatic popup matters.

13. Risks and mitigations

Risk	Impact	Mitigation
Automatic popup feels intrusive.	Users disable or uninstall PromoBase.	One popup per merchant visit, easy dismissal, and observe behavior in prototype sessions. The popup is opt-in per site, waits 10 minutes between reminders, and offers "Don't remind me again" (D-004, D-005, D-064).
Site permission prompts cause drop-off.	Reminders are not enabled for saved records.	Request access only when ~~Notify is enabled~~ the user turns reminders on; explain the reason and support later opt-in. The yellow icon works without any access, so a declined dialog still leaves a visible cue (D-061).
Domain parsing creates false matches.	Wrong codes appear on unrelated sites or expected reminders fail.	Use Public Suffix List-aware parsing and acceptance cases for subdomains and deceptive suffixes.

Risk	Impact	Mitigation
Guest records are lost after extension removal or profile loss.	User loses saved codes.	Persistent warning; document cloud account and export/import as future recovery options. Export and import ship in the first release (D-106, D-108).
Codes are expired or restricted by merchant terms.	User trusts a code that fails at checkout.	Keep expiry warnings visible and state that PromoBase does not verify code validity. Said in the welcome and in the library (D-113).
Single developer maintenance burden.	Security and Chrome API changes can break reminders.	Keep permissions narrow, dependencies limited, and test the supported Chrome range before releases. Monthly dependency updates with a cooldown, security updates as they come, browser checks on every change, and a written release procedure (D-121, D-129).
Saved data from a newer or damaged version.	The library cannot show the codes.	The data is reported and left untouched, with a copy to download and a way to report it (D-042, D-124).

14. Delivery approach

A staged build reduces unknowns and keeps the first deliverable testable before implementing the full extension workflow.

Stage	Deliverable	Exit condition
1. Product skeleton	Manifest V3 extension, popup shell, local record schema.	Loads unpacked in Chrome and can display the popup. Done: PR #1.
2. Local record management	Create, list, edit, delete, validation, capacity handling.	All record operations pass acceptance criteria using local storage. Done together with stage 3, after the stage 4 spike: PRs #3 to #5 (D-028, D-068).
3. Domain matching	Normalization utility and domain test cases.	Root, www, subdomain, public suffix, and deceptive-domain cases resolve correctly.
4. Reminder proof	Per-domain permission, ~~dynamic script,~~ action popup and deduplication. Plus the icon highlight.	One correct popup appears on an allowed matching visit and not on a non-matching site. Done: spike PR #2, then PRs #6 and #7 (D-091, D-099).
5. Polish and publish prep	Accessibility, empty/error states, privacy copy, Web Store assets. Plus export and import, and browser checks in CI.	Manual QA and listing disclosures are ready for review. Done: PRs #8 to #26, ending with the release procedure (D-105).

15. Future improvements

The following ideas are intentionally optional and low priority. They should be revisited after the local save-and-remind workflow has been tried with real users.

15.1 Account and durable storage

Allow email-based account creation and synchronize records to a backend so records can survive extension removal, profile reset, or use on another device. Include export, account deletion, and recovery behavior. Candidate stack: Python FastAPI, PostgreSQL, and a managed email identity provider.

15.2 Sharing

Allow a user to share selected codes with another PromoBase user by email address or PromoBase user ID. Define recipient confirmation, revocation, duplicate behavior, sender/recipient ownership, and audit history before implementation.

15.3 Monetization and support

Consider donations and a paid plan after ongoing value is established. Potential subscription benefit: an ad-free experience. Payment processor, cancellation flow, refunds, taxes, and store policy requirements remain undecided. Avoid embedded ads in the first release.

15.4 Product enhancements

- Source context such as creator name, video URL, platform, and date discovered.
- Optional tags, categories, or redeemed status.
- ~~Export and import for local users before cloud accounts exist.~~ Moved into the first release (PB-016).
- Reminder style settings, including toolbar popup, in-page card, and standard system notification where supported.
- Support for other browsers after the Chrome experience is stable.
- Other interface languages and date formats; English and DD.MM.YYYY only in the first release (D-034).

15.5 User feedback

Consider a small Provide Feedback action with an optional five-star rating and free-form comment to support early product learning. Defer direct email submission until there is a secure and privacy-transparent delivery method. A public extension package must not contain email-service credentials. A future implementation should state what information will be sent, ask for consent before submission, send only the rating and comment (plus any explicitly optional diagnostic details), protect the endpoint against spam, and identify the receiving support address or service in the privacy notice. A hosted feedback form can be evaluated as a lower-maintenance alternative to operating a custom email backend.

16. Decisions and assumptions

Topic	Initial decision
Account	No account required for first release.
Storage warning	Records are local to the Chrome profile; extension removal or profile loss may delete them.
Record cap	100 records for guest users.
Record entry	Manual entry only. Or import of a PromoBase export (D-108).
Use case	General coupon organizer with emphasis on creator codes personally encountered.
Domain matching	Use registrable domain; root and all subdomains match.
Reminder surface	Open PromoBase's Chrome toolbar popup, not a system notification, for the initial prototype. Plus the yellow toolbar icon on every saved site; the popup opens by itself only on sites with reminders on (D-061, D-064).
Dates	Date-only values. Expiry is inclusive to the end of the date in the current local timezone.
Expired records	Retained, grayed out, sorted below active records, and still shown with a warning on visit.

Topic	Initial decision
Undated records	Remain in the extension indefinitely.
Stack	TypeScript, Manifest V3, React/Vite, chrome.storage.local. Python/FastAPI/PostgreSQL later.
Market score	4/10 qualitative hypothesis pending user validation.
Open source	Public GitHub repository under MIT. Add the license file before public release.
Date format	Display and accept DD.MM.YYYY. Persist ISO YYYY-MM-DD internally.
Notify default / reminders	~~Checked only when a usable Resource or Product URL exists; otherwise unchecked and disabled.~~ Off by default; offered after a save and switched per site, for all its codes (D-064, D-100).
Save failures	Preserve input, explain the issue, and never claim a failed write succeeded.
Feedback form	Potential low-priority follow-up. Requires consent, privacy disclosure, and secure delivery.
Release	Tag-triggered package checked by CI, uploaded to the store by hand; no store credentials in CI (D-121).

16.1 Assumptions to keep visible

- Start Date means the code should not generate a reminder before that local calendar date.
- A record with Resource or Product URL left empty can still be saved, but cannot trigger a site-visit reminder.
- If several codes match one merchant, PromoBase displays them together in a single popup.
- Codes are user-provided and are not verified by PromoBase. Merchant restrictions may apply.
- Chrome 127 or later is the minimum proposed version for programmatic action popup support.

17. Glossary

Term	Meaning
Promo code	A text code a user can enter at a merchant to request a discount or promotion.
Merchant domain	The registrable domain used to associate a record with a merchant website, such as dropbox.com.
Registrable domain	The domain a person can register under a public suffix, derived with Public Suffix List rules.
Host or hostname	The host portion of a URL, such as www.dropbox.com or shop.dropbox.com.
Manifest V3	The current Chrome extension architecture and manifest format used for the proposed build.
Optional host permission	A site access permission that an extension can request later for a specific host rather than requesting access to all sites at install time.
Guest mode	Using PromoBase without an account, with records stored in the current browser profile. A term for this document; the interface does not show it (D-128).
Reminders on a site	The per-site switch that lets PromoBase open its popup by itself on a visit; it needs site access (D-100).

Term	Meaning
Icon highlight	The yellow toolbar icon on a site with saved codes, changed by Chrome with no site access (D-061, D-093).

18. Additional Notes

How to read this version. v0.3 was the entry PRD, kept unchanged during development; the decisions log recorded each change and its reason (D-065). This version applies them:

- Struck-through text was in v0.3 and was dropped.
- Text highlighted in green was added or changed, with the decision behind it in brackets, for example (D-064).
- Everything else is v0.3 as it was, and was built as written.

The decisions log is published as docs/decisions.md in the repository. It covers D-001 to D-131; this version reflects them up to the release procedure, and the log stays the place for the reasons behind each change.

Release and maintenance, beyond v0.3: RELEASING.md describes how a version reaches the Chrome Web Store (D-121); CONTRIBUTING.md the rules for changing stored data and reviewing dependency updates (D-119, D-129); the icons, tile and store screenshots are made by scripts so that they carry no metadata (D-114, D-117).

Fixed from v0.3: section 18 was empty; R15 repeated R12 and is merged into it; R11 is now referenced in 9.3.

19. References

Platform behavior and competitor descriptions were checked against the following official documentation and product help pages on 8 October 2026. Product feature descriptions may change over time. R16 and R17 were added on 11 November 2026.

R1. Chrome for Developers. chrome.storage API. Open source

R2. Chrome for Developers. Storage and cookies in extensions. Open source

R3. Chrome Web Store. User data privacy and secure handling requirements. Open source

R4. Chrome for Developers. Permissions API and optional permissions. Open source

R5. Chrome for Developers. Action API. Open source

R6. Chrome for Developers. Add a popup. Open source

R7. Chrome for Developers. Content scripts. Open source

R8. PayPal Honey Help. What is the Honey extension. Open source

R9. Capital One Shopping Help Center. Open source

R10. Rakuten Help. Browser Extension Help. Open source

R11. Chrome for Developers. Manifest file format. Open source

R12. Chrome Web Store. Program policies. Open source

R13. Public Suffix List. Open source

R14. GitHub Docs. Licensing a repository. Open source

~~R15. Chrome Web Store. Program policies. Open source~~ Same as R12.

R16. Chrome for Developers. chrome.declarativeContent API. Open source

R17. Chrome for Developers. The activeTab permission. Open source